Dylan Hensley
ECEN 164 HW4

1. Buck reg steps down 60V to 12V

Output power required = 50W, $V_{REF} = 5V$

Part 1

a. $\frac{50}{12} = \boxed{4.167A}$

b. $eff = \frac{V_{out}}{V_{in}} = \frac{12}{60} = \boxed{20\%}$ yikes

c. $P_{in} = 60(4.167) = 250W$

$P_{out} = 12(4.167) = 50W$

$250 - 50 = \boxed{200W \text{ lost}}$

d.

$200W = 200 J/s$ 1 year has

$365 * 24 * 60 * 60$

$= 31536000s$ in

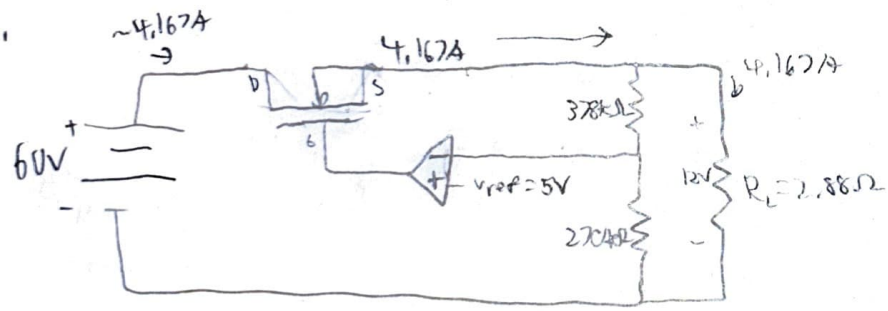
$200 * 31536000 = \boxed{6.3072 \times 10^9 J}$

e. $kWh = kJ/s \cdot h = kJ \cdot (3600) = J \cdot 3.6 \times 10^6$

$\frac{6.3072 \times 10^9}{3.6 \times 10^6} = 1752$ | $1752(.1) = \boxed{\$175.20}$

Part 2

a.



$\frac{R_{f2}}{R_{f1} + R_{f2}}(12) = 5$

$R_{f1} = 270k$ $\frac{R_{f2}}{R_{f1} + R_{f2}} = \frac{5}{12} = \frac{R_{f2}}{270k + R_{f2}} \rightarrow$ $1350 = 7R_{f2}$

$R_{f2} = 192.857$

vs

$R_{f2} = 270k$ $\frac{270}{R_{f1} + 270} = \frac{5}{12} \rightarrow$ $5R_{f1} + 1350 = 3240$

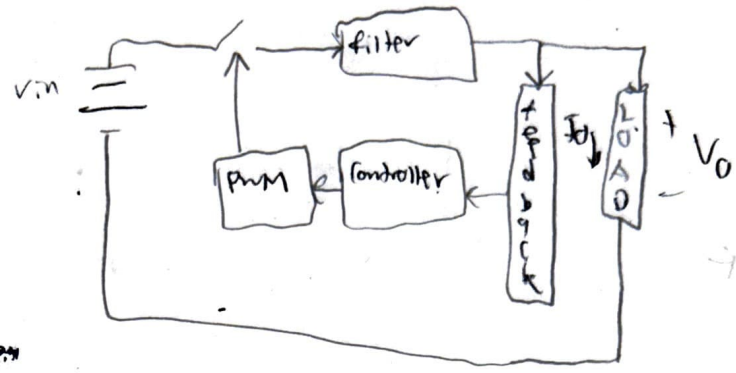
$R_{f1} = 378\Omega$

I choose this

b. There is no ground current b/c we are using a MOS device (no gate current).

2.

Buck regulator



IGNORE

A current source $i(t) = 3 + 4\cos(2\pi 60t) + 6\cos(4\pi 60t)$ A is connected to a parallel RC load with $R = 100\ \Omega$ and $C = 50\ \mu F$. Determine the average power absorbed by the load.

a. The converter uses a switch to rapidly turn on and off a DC voltage. The LC filter store and release energy smoothing the output. The feedback portion allows the controller to compare against a lower voltage (eff improvement) and the controller stabilizes differences in voltage.

b.

PWM

in: controller out

out: switch (MOS or BJT)

func: set the switching duty cycle

Filter

in: Vin (when switch closed)

out: smoothed Vin

func: smooths input to actually make it more DC on the out

Controller

in: $V_{feedback}$ and V_{ref}

out: voltage to PWM corrected

func: correct and stabilize voltage to load

feedback

in: V_o

out: voltage divided V_o (close to Vref)

func: give inputs to controller, voltage divider circuit helps improve eff.

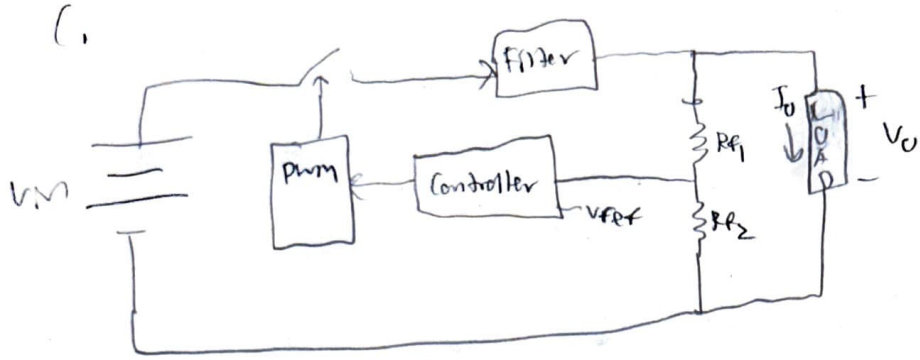
Load

in: V_o, I_o

out: I_o

func: dissipate power

c.



Changing Vref will change PWM duty cycle which will change Vo and Io, thus setting a new Vo

d. When Vin drops, assuming its still above Vo at all thresholds, Vo will stay the same. Vo is only dependant on Vref and the voltage divider uses Vo. If Vin drops, The PWM will correct to the duty cycle so Vo stays constant.